William Richards

Senior Front-End Engineer

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SUMMARY

Software Engineer with 9 years of experience, including 7 years specializing in front-end development and 5 years building enterprise-scale web applications for Fortune 500 clients. Led front-end architecture on an AI governance platform at ModelOp, expanding scope into Python SDK development, test automation infrastructure, and cloud deployments. Deep expertise in Angular and TypeScript; actively working in React for modern full-stack projects.

EXPERIENCE

Senior Front-End Engineer | **ModelOp** *Apr 2021 – Present*

- Owned front-end architecture end-to-end for an AI governance platform serving Fortune 100 clients, leading feature delivery across Angular 17+, TypeScript, RxJS, NgRx, and SCSS
- Drove a performance and accessibility overhaul: achieved WCAG 2.2 AA compliance and 320px responsiveness while cutting initial load time 40% and bundle size 60% via Lazy Loading and Tree Shaking.
- Introduced Tailwind CSS and a token-based design system (SCSS + Figma Tokens Studio) across a 5-engineer front-end team, standardizing styling conventions and eliminating a class of style-related bugs.
- Built a test automation SDK in Python abstracting common UI workflows into reusable, composable functions, enabling scalable pytest/Selenium test authoring; automated 20 of 400 manual regression tests as an initial implementation
- Contributed Spring Data JPA repository methods using derived query conventions; collaborated across Docker and Kubernetes environments to diagnose and resolve production issues alongside Java team.
- Refactored change detection to a zoneless, signal-based architecture, eliminating Zone.js overhead and reducing unnecessary CD cycles; measured runtime improvements via Lighthouse, yielding meaningful reductions in Total Blocking Time (TBT).
- Drove adoption of Chart.js, Angular 17, and AI development tooling by presenting technical roadmaps to the Director of UI/UX, VP of Product, and CTO.

Web Developer | **University of Utah – Dept. of Human Genetics** *May 2019 – Dec 2020*

- Built a genomic pedigree visualization platform with Vue, D3, and Node.js; leveraged Vuex, Web Workers, and OffscreenCanvas for high-interactivity rendering.
- Deployed in a secure AWS environment using OAuth 2.0, CloudFormation, and IAM policies for consistent, auditable provisioning.

Software Developer | **University of Utah – Dept. of Human Genetics** *Jun 2017 – Apr 2019*

- Refactored genomic variant-detection pipelines in C++11/CMake, achieving a 10× build-time reduction; built high-throughput ALU detection tools using FASTAHACK, MINIMAP2, and lock-free hash tables.
- Trained TensorFlow CNN classifiers to detect tissue contamination in xenograft genomic data, achieving 92% accuracy for downstream oncological research.

PROJECTS

Blotify | React, Next.js | Automated western blot and cytokine array quantification; deployed and actively used as a production research tool at a university lab | blotify.dev

TECHNICAL SKILLS

Languages	TypeScript, JavaScript (ES6+), Python 3, C++11
Frameworks	Angular 17+, React, Vue, Node.js, RxJS, NgRx, Redux Toolkit
Styling	SCSS, Tailwind CSS, Material Design, Design Tokens, Figma
Testing	Vitest, Jasmine, Jest, Selenium, Playwright, Lighthouse, Axe
Tools & Cloud	Git, Webpack, Vite, Docker, AWS, ESLint, D3.js
Standards	WCAG 2.2 AA, Atomic Design, Component-Driven Dev, Agile / Scrum

EDUCATION

Bachelor of Arts, Computer Science | Carleton College, Northfield, MN *2013 – 2017*

AWS Certified Cloud Practitioner | **AWS Certified AI Practitioner** · *2026 – 2029*